Nima Hekmati Farid

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Research Interests

- Core Architecture: Computer Architecture, Multi-GPU & Heterogeneous Systems, Memory Hierarchies, Cache Coherence.
- Systems & Modeling: Interconnect Bottlenecks, NUMA Data Locality, Parallel Computing, GPU Architecture Modeling.

Education

Sharif University of Technology — Tehran, Iran

B.Sc. in Computer Engineering – Major GPA: 18.54 / 20.0 (Cumulative: 18.03 / 20.0) — Sep 2023 – Expected Spring 2028

- Architecture & Systems (GPA: 18.96/20): Computer Structure & Machine Language (20/20), Digital Systems Design (19.1/20), Multicore Computing (18.8/20), Embedded Systems (18.5/20), Computer Architecture (18.4/20).
- Foundations & Current Courses: Data Structures & Algorithms (20/20), Design of Algorithms (19.5/20) | In Progress: Operating Systems, Computer Networks, Real-Time Systems.

Honors & Awards

- National University Entrance Exam (Konkur): Ranked 81st nationwide among 150,000+ candidates in Mathematics & Physics (Top 0.05%) — 2023

Research Experience

Undergraduate Researcher – Multi-GPU Systems — May 2026 – Present

Sharif University of Technology (Supervised by Prof. Hamid Sarbazi-Azad) — Tehran, Iran

- Investigating memory hierarchy bottlenecks in multi-GPU architectures, focusing on cache coherence, NUMA data locality, and interconnect traffic under high memory contention.
- Configuring and evaluating architectural mechanisms using the Multi-GPU architectural simulator (MGPUSim), analyzing compute unit execution, memory controller contention, and interconnect latencies.
- Conducted in-depth analysis of 20+ research papers across leading architecture venues (ISCA, MICRO, HPCA) to identify evaluation baselines and open challenges in multi-GPU scaling.

Teaching Experience

Teaching Assistant – Computer Structure and Machine Language — Fall 2026

Sharif University of Technology (Instructor: Prof. Hamid Sarbazi-Azad) — Tehran, Iran

- Designed practical assignments covering assembly programming, CPU datapath, and instruction decoding.
- Developed automated grading testbenches to evaluate student code for functional correctness and edge cases.
- Assisting in designing comprehensive course projects and holding technical consultation hours for students.

Selected Projects

- 16-Bit Stack-Based CPU: Designed and implemented a custom processor in Logisim, including a stack-based datapath, ALU, microprogrammed control unit, instruction decoding, and conditional branching.
- Gigit (Lightweight Version Control System): Implemented a Git-like version control system in C and Bash for Unix systems, supporting content-addressable blob storage, tree objects, commit graphs, and branch management. [GitHub]
- Pet Tracker (IoT System): Developed telemetry and communication modules for an embedded pet tracking system using GPS telemetry and MQTT for real-time localization and activity monitoring. [GitHub]
- Price Tracker Pro: Built a distributed price-tracking backend using FastAPI, Kafka, Celery, and PostgreSQL for concurrent scraping and event processing. [GitHub]

Work Experience

Backend Developer — Feb 2025 – Present

Mahsan Company — Tehran, Iran

- Profiled and optimized backend microservice endpoints, achieving a 20% reduction in response latency through database query optimization, indexing, and serialization improvements.
- Engineered high-throughput microservices using FastAPI and Django, orchestrating distributed asynchronous workflows with Temporal and Kafka.
- Managed containerized deployment environments utilizing Docker, Kubernetes, and relational databases.

Technical Skills

Systems & Languages: C, Python, x86 Assembly, MIPS Assembly, Verilog, Java, Bash, SQL

Architecture & Simulation Tools: MGPUSim, Logisim, ModelSim, Proteus, Git, Linux/POSIX

Infrastructure & Backend: Docker, Kubernetes, Kafka, Temporal, Celery, FastAPI, Django, PostgreSQL, Redis